

Cai Chen

achen2304@gmail.com | (641) 808-8022 | Ames, IA | LinkedIn/Achen2304 | GitHub/Achen2304

Education

Iowa State University

Bachelor of Science in Computer Engineering

Expected Graduation: May 2027

GPA: 3.5

Projects

CharDle.com – Daily Character Guessing Game

- Spearheaded the development of a **React**-based frontend with **multiple developers** using **Vite**, implementing core features with **React Router** for navigation, **React Context API** for state management, and **fetch API** for **RESTful** endpoint integration that drives **300+ daily active users**.
- Crafted responsive UI components using **Tailwind CSS/Daisy UI** and **React Icons**, enhancing user experience with **React-Hot-Toast** notifications and dynamic feedback systems across **all device sizes**.
- Implemented robust **data persistence** using **local storage** and **js-cookie** for game progress tracking, and **time zone-aware game state management**, ensuring regular daily resets and gameplay experiences for users across **40+ different countries**.
- Engineered a scalable **Node.js/Express** backend with **MongoDB/Mongoose** integration, designing optimized **schemas** for character data, while implementing **RESTful APIs** that serves **1000+ unique character entries** and maintains **99.9% uptime**.
- Coordinated development through **GitHub project boards**, managing and assigning feature implementations and bug fixes while collaborating with team members using Discord for communication.

SOLAR Project – ISU Swan Hacks 2024

- Developed **Spring Boot** backend for an educational platform in a team of 3, implementing **user authentication** and classroom management system with **JPA/Hibernate** entities supporting teacher/student role **hierarchies** and classroom **relationships**.
- Created **RESTful API** endpoints and controllers specifically designed for Unity game integration, handling game data submissions and progress tracking with custom data structures for planet exploration metrics.

Caizhen.com – Portfolio Website

- Architected a modern portfolio website using **React + Vite**, implementing smooth page transitions with **React Router** and **AOS** library, while ensuring optimal loading performance through **lazy loading**.
- Developed an intuitive UI featuring dynamic hover states and **mobile-responsive** navigation, utilizing **CSS modules** and a consistent design system through **reusable components**.
- Integrated a **full-stack** contact system using **Node.js/Express** backend with **Nodemailer** for automated email notifications, alongside a responsive PDF resume viewer component using **react-pdf-viewer**, ensuring secure communication through environment variables.

Skills and Certifications

Languages: JavaScript, HTML, CSS, Java, C, Python

Technologies: React.js, Node.js, Express.js, Git, GitHub, Tailwind CSS, RESTful APIs, MongoDB, Mongoose, SQL, Bash

Free Code Camp Certifications: Back End Development and APIs Certification (Completed Jan 2025), JavaScript Algorithms and Data Structures Certification (Completed Feb 2025)

Experience

Independent Computer Repair

December 2024 – Present

Part Time – Per Customer Request

Ames, IA

- Diagnosed and repaired diverse **hardware/software issues** for laptops and PCs, achieving **100% repair** success rate through **component assembly/disassembly** and comprehensive **troubleshooting** techniques.